

THE ZERO PARADOX

ZP-L: Incomputability Convergence

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v1.0: Initial release. All theorems §I–§VII proved sorry-free in Lean 4. Axiom footprint: [propext, Classical.choice, Quot.sound] throughout.

ZP-L establishes four components of the incomputability convergence argument. First, Classical.choice appears at the non-constructive diagonal step in each of the four ZP formal languages — topology, information theory, set theory, and computation. Second, Roger’s fixed-point theorem (Kleene’s second recursion theorem) is formalized as a wrapper, grounding the computational fixed-point structure. Third, the ordinal ϵ_0 is fully characterized as the first fixed point of $\alpha \mapsto \omega^\alpha$ and the limit of the tower $\omega, \omega^\omega, \omega^{\omega^\omega}, \dots$. Fourth, ordinals below ϵ_0 encode into $\mathbb{Z}_2$ via their Cantor normal form, and as the tower stages approach ϵ_0 , their 2-adic encodings converge to $0 = \perp$.

The central result is the canonical snap closure (§VII): the map $\phi \alpha = \text{if } \alpha < \epsilon_0 \text{ then } c_0 \text{ else } c_1$ simultaneously satisfies all five conditions — monotone, tower-aligned, fixed-point-respecting, snapping at ϵ_0 , and ϵ_0 minimal. All conditions are verified without free hypotheses for this witness. 24 theorems proved, zero sorry, axiom footprint [propext, Classical.choice, Quot.sound] throughout.

Section I: Axiom Footprint Convergence

Non-constructibility appears in four formal languages across the ZP framework. In each proved layer, Classical.choice is required at the diagonal step where a non-constructive choice is forced. This is not a contingent feature of the proofs — it is the formal expression of the claim that the diagonal argument cannot be eliminated.

Layer	Formal Language	Expression of non-constructibility
ZPB	Topology	C3: no continuous path $\perp \rightarrow x \neq \perp$
ZPC	Information Theory	L-INF: infinite surprisal at $\perp$
ZPJ/K	Set Theory + Computation	bot_self_mem (AFA); botCode (Kleene)
ZPI	Algorithmic IT	$K(S^{\mathbb{Z}_2} n)/ S^{\mathbb{Z}_2} \rightarrow 1$; K uncomputable

Remark: Why K is Absent from Lean

Kolmogorov complexity K is not computed in Lean in this framework. Its existence as a total function requires Classical.choice — exactly the axiom Nat.Partrec.Code.fixed_point₂ already uses in ZP-K. The AFA/Kleene route reaches the same fixed-point structure via a provable path without requiring K to be explicitly computed.

Axiom footprint evidence — the following ZP-K theorems all carry [propext, Classical.choice, Quot.sound]:

Remark: Why K is Absent from Lean

ZPK.t_comp (T-COMP four-way equivalence)

ZPK.da1_paths_unified

ZPK.isComputationalQuine_undecidable

ZPK.infinite_quine_family

The Classical.choice entry is the computational expression of the diagonal. ZP-L inherits this footprint throughout.

Section II: Roger Fixed-Point Stability

Roger's fixed-point theorem (also known as Kleene's second recursion theorem) states that any computable transformation of a code has a behavioral fixed point: a code c such that running $f(c)$ and running c produce the same partial function. The fixed-point structure is stable under any computable transformation.

Theorem: roger_fixed_point_stability (ZPL.lean §II)

For any computable $f : \text{Code} \rightarrow \text{Code}$,

$\exists c : \text{Code}, \text{eval } (f\ c) = \text{eval } c$

Proof: wrapper around ZPK.roger_fixed_point_exists.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Note: this is an existential result. The specific code c is not constructively produced; Classical.choice selects it. This is the same non-constructive step that appears across all ZP layers (§I).

Section III: The Ordinal ϵ_0

The ordinal ϵ_0 is the smallest fixed point of the map $\alpha \mapsto \omega^\alpha$. It is the supremum of the tower $0, 1, \omega, \omega^\omega, \omega^{\omega^\omega}, \dots$ — each stage is strictly below ϵ_0 , and ϵ_0 is not reached by any finite iteration. This section is entirely within Lean scope via Mathlib's ordinal machinery.

Definitions (ZPL.lean §III)

$\text{epsilonZero} : \text{Ordinal} := \text{Ordinal.epsilon } 0 (= \text{nfp } (\omega^\cdot) 0 = \text{veblen } 1\ 0)$

$\text{fundamentalSeq} : \mathbb{N} \rightarrow \text{Ordinal} := \text{fun } n \mapsto (\alpha \mapsto \omega^\alpha)^\wedge[n]\ 0$

Explicit stages:

$\text{fundamentalSeq } 0 = 0$

$\text{fundamentalSeq } 1 = 1$

Definitions (ZPL.lean §III)

$\text{fundamentalSeq } 2 = \omega$

$\text{fundamentalSeq } 3 = \omega^\omega$

$\text{fundamentalSeq } (n+1) = \omega^{(\text{fundamentalSeq } n)}$

Theorem: epsilonZero_fixedPoint

$\omega^{\varepsilon_0} = \varepsilon_0$

Proof: Ordinal.omega0_opow_epsilon 0 (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: epsilonZero_eq_nfp

$\varepsilon_0 = \text{nfp } (\omega^{\cdot}) 0$

Proof: Ordinal.epsilon_zero_eq_nfp (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: epsilonZero_eq_iSup

$\varepsilon_0 = \sqcup n : \mathbb{N}, \text{fundamentalSeq } n$

Proof: iSup_iterate_eq_nfp (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: epsilonZero_tower_lt

$\forall n : \mathbb{N}, \text{fundamentalSeq } n < \varepsilon_0$

Every finite stage of the tower is strictly below ε_0 .

Proof: Ordinal.iterate_omega0_opow_lt_epsilon_zero (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: epsilonZero_le_fixedPoint

$\forall b : \text{Ordinal}, \omega^b = b \rightarrow \varepsilon_0 \leq b$

ε_0 is the least fixed point of $\omega^{\cdot}$ above 0.

Proof: Ordinal.epsilon_zero_le_of_omega0_opow_le (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: fundamentalSeq_strictMono

$\forall n : \mathbb{N}, \text{fundamentalSeq } n < \text{fundamentalSeq } (n + 1)$

The tower is strictly monotone.

Proof: by induction; successor step uses `isNormal_opow`.

Lean purity: `[propext, Classical.choice, Quot.sound]`. ✓

Remark R-L.1: Proof-Theoretic Alignment

Gentzen's theorem (1936) establishes that ϵ_0 is the proof-theoretic ordinal of Peano Arithmetic: PA can prove transfinite induction for any ordinal strictly below ϵ_0 , but not for ϵ_0 itself. This is not claimed or proved here.

ZP-L derives ϵ_0 as the snap threshold from ordinal fixed-point structure, independently of proof theory. The coincidence with PA's proof-theoretic ordinal is a structural alignment: both derivations locate the same boundary — the ordinal where ω -tower self-iteration becomes self-limiting. Two independent formal derivations converging on the same ordinal is corroborating evidence that ϵ_0 is the structurally correct threshold, not an arbitrary choice.

Section IV: Cantor Normal Form Bridge

Every ordinal below ϵ_0 has a unique Cantor normal form — a finite expression $a_1 \cdot \omega^{e_1} + a_2 \cdot \omega^{e_2} + \dots + a_k \cdot \omega^{e_k}$ with $e_1 > e_2 > \dots > e_k$ and $a_i < \omega$. In Lean: `NONote` (`Mathlib.SetTheory.Ordinal.Notation`). The encoding `cnfToZp2` maps each such ordinal to $\mathbb{Z}_2$ via structural recursion on the Cantor normal form.

Definition: cnfToZp2 (ZPL.lean §IV)

`cnfToZp2 : NONote →  $\mathbb{Z}_2$`

Base: `cnfToZp2(0) = 0`

Recursive: `cnfToZp2( $\omega^e \cdot n + a$ ) =  $2^{(v_2(\text{cnfToZp2}(e)) + 1)} \cdot n + \text{cnfToZp2}(a)$`

where v_2 denotes the 2-adic valuation.

Valuation of the n -th tower stage:

`cnfToZp2(towerNONote 0) = 0` (valuation 0)

`cnfToZp2(towerNONote 1) = 2` (valuation 1)

`cnfToZp2(towerNONote 2) = 4` (valuation 2)

`cnfToZp2(towerNONote n) =  $2^n$` (valuation n)

Theorem: cnfToZp2_tower_valuation

$\forall n : \mathbb{N}, (\text{cnfToZp2 } (\text{towerNONote } n)).\text{valuation} = n$

Theorem: <code>cnfToZp2_tower_valuation</code>
The 2-adic valuation of the n -th tower stage encoding equals n .
Proof: by induction; successor step uses <code>PadicInt.valuation_pow</code> .
Lean purity: <code>[propext, Classical.choice, Quot.sound]</code> . ✓

Theorem: <code>cnfToZp2_valuation_unbounded</code>
$\forall k : \mathbb{N}, \exists \alpha : \text{NONote}, k \leq (\text{cnfToZp2 } \alpha).\text{valuation}$
The 2-adic valuation is unbounded across ordinals below ε_0 .
Proof: <code>towerNONote k</code> witnesses the bound.
Lean purity: <code>[propext, Classical.choice, Quot.sound]</code> . ✓

Theorem: <code>tower_converges_to_zero</code>
<code>Filter.Tendsto (fun n ↦ cnfToZp2 (towerNONote n)) Filter.atTop (nhds 0)</code>
The tower encodings converge to $0 = \perp$ in $\mathbb{Z}_2$.
Proof: each stage <code>cnfToZp2(towerNONote (n+1)) = 2^{^(n+1)}</code> in $\mathbb{Z}_2$, so its 2-adic norm is $\ 2\ ^{(n+1)} = (1/2)^{(n+1)} \rightarrow 0$. Uses <code>Metric.tendsto_atTop</code> and <code>exists_pow_lt_of_lt_one</code> .
Lean purity: <code>[propext, Classical.choice, Quot.sound]</code> . ✓

Section V: Ordinal Tower Limit and Snap Threshold

This section connects the ordinal tower to the ZPE machine-phase snap. The cofinality theorem shows that the fundamental sequence approaches ε_0 from below. The lower-bound theorem shows that any monotone tower-aligned map must assign c_0 to all pre- ε_0 ordinals. A canonical witness snapping exactly at ε_0 is exhibited.

What this does NOT claim (§V)
Gentzen’s theorem: that ε_0 is the proof-theoretic ordinal of PA (not claimed — see Remark R-L.1).
Any statement about formal provability in PA.
That ε_0 is the UNIQUE minimal snap boundary: <code>snap_threshold_is_epsilon_zero</code> shows no ordinal below ε_0 works (for maps satisfying the stated hypotheses), but does not rule out maps that snap at some ordinal strictly above ε_0 .
That the snap threshold result applies to all maps <code>Ordinal → MachinePhase</code> regardless of the monotonicity and tower-alignment hypotheses.

Theorem: fundamentalSeq_cofinal

$\forall \alpha : \text{Ordinal}, \alpha < \varepsilon_0 \rightarrow \exists n : \mathbb{N}, \alpha < \text{fundamentalSeq } n$

The fundamental sequence is cofinal in ε_0 : for any ordinal below ε_0 , some tower stage exceeds it.

Proof: $\varepsilon_0 = \text{nfp } (\omega^\cdot) 0$ (`epsilonZero_eq_nfp`), and `lt_nfp_iff` gives $a < \text{nfp } f \ b \leftrightarrow \exists n, a < f^n[n] \ b$.

Lean purity: [`propext`, `Classical.choice`, `Quot.sound`]. ✓

Theorem: snap_threshold_is_epsilon_zero

For any $\phi : \text{Ordinal} \rightarrow \text{MachinePhase}$ satisfying:

(a) `hmono`: $\forall \alpha \leq \beta, \text{join } (\phi \ \alpha) (\phi \ \beta) = \phi \ \beta$ (order-non-decreasing in the `ZPSemilattice` sense: $\phi \ \alpha \leq \phi \ \beta$)

(b) `h0`: $\forall n : \mathbb{N}, \phi \ (\text{fundamentalSeq } n) = c_0$

we have: $\forall \alpha < \varepsilon_0, \phi \ \alpha = c_0$

This is a lower bound: no ordinal below ε_0 is a snap point for maps satisfying (a) and (b). "Minimal" not "unique."

Proof: cofinality gives a stage above α ; monotonicity chains $\phi \ \alpha \leq \phi(\text{stage}) = c_0$; $c_0 = \perp$ forces $\phi \ \alpha = c_0$.

Lean purity: [`propext`, `Classical.choice`, `Quot.sound`]. ✓

Theorem: c1_epsilon_zero_identification

$\exists \phi : \text{Ordinal} \rightarrow \text{MachinePhase},$

$(\forall n : \mathbb{N}, \phi \ (\text{fundamentalSeq } n) = c_0) \wedge \phi \ \varepsilon_0 = c_1$

Witness: $\phi \ \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$.

One specific map sends all tower stages to c_0 and ε_0 to c_1 .

The stronger structural claim — an order-preserving morphism $\text{Ordinal} \rightarrow \mathbb{Z}_2$ `MachinePhase` compatible with the `CNF` $\rightarrow \mathbb{Z}_2$ encoding — remains outside Lean scope: no type bridge between `Ordinal` and `MachinePhase` is defined in this library.

Lean purity: [`propext`, `Classical.choice`, `Quot.sound`]. ✓

Section VI: Kleene-Ordinal Fixed-Point Bridge

The ordinal fixed-point structure ($\varepsilon_0 = \text{nfp } (\omega^\cdot) 0$, $\omega^\varepsilon_0 = \varepsilon_0$) and the computational fixed-point structure (Kleene's recursion theorem, `roger_fixed_point_stability`) both require `Classical.choice` at their non-constructive step — parallel structure, not a proved isomorphism. The hypothesis `hfp` encodes that ordinal fixed points of $\omega^\cdot$ map to the snap state c_1 . Under this hypothesis plus monotonicity and tower alignment, ε_0 is the minimal snap threshold.

Theorem: snap_exactly_at_epsilon_zero

For any $\phi : \text{Ordinal} \rightarrow \text{MachinePhase}$ satisfying:

(a) hmono: order-non-decreasing ($\text{join } (\phi \alpha) (\phi \beta) = \phi \beta$ for $\alpha \leq \beta$)

(b) h0: every tower stage maps to c_0

(c) hfp: every fixed point of $\omega^\cdot$ maps to c_1

we have: $\phi \varepsilon_0 = c_1$ AND $\forall \alpha, \phi \alpha = c_1 \rightarrow \varepsilon_0 \leq \alpha$

ε_0 is the minimal snap threshold: ϕ assigns c_1 first at ε_0 .

Note: "minimal" not "unique." A map satisfying these hypotheses could also assign c_1 to ordinals above ε_0 ; what is ruled out is any snap strictly before ε_0 .

Proof: upper bound from hfp + epsilonZero_fixedPoint; lower bound from snap_threshold_is_epsilon_zero.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: kleene_ordinal_snap_bridge

$\exists \phi : \text{Ordinal} \rightarrow \text{MachinePhase},$

$(\forall n : \mathbb{N}, \phi (\text{fundamentalSeq } n) = c_0) \wedge$

$(\forall \alpha, \omega^\alpha \alpha = \alpha \rightarrow \phi \alpha = c_1) \wedge$

$\phi \varepsilon_0 = c_1 \wedge$

$\forall \alpha, \phi \alpha = c_1 \rightarrow \varepsilon_0 \leq \alpha$

Witness: $\phi \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$.

Note: the theorem name refers to the informal §VI conceptual parallel between Kleene recursion fixed points and ordinal fixed points of $\omega^\cdot$. The theorem itself is purely ordinal — no Code or eval appears.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Section VII: Canonical Snap Map — Full Closure

The canonical threshold map $\phi \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$ satisfies all three conditions of snap_exactly_at_epsilon_zero (hmono, h0, hfp). For this specific map, the snap identification is unconditional: all five conditions are simultaneously verified without free hypotheses.

Theorem: snap_map_mono

$\forall \alpha \beta : \text{Ordinal}, \alpha \leq \beta \rightarrow$

$\text{join } (\text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1)$

Theorem: snap_map_mono

(if $\beta < \varepsilon_0$ then c_0 else c_1) =

if $\beta < \varepsilon_0$ then c_0 else c_1

The canonical map is order-non-decreasing.

Proof: four cases (α/β vs ε_0). The case $\alpha \geq \varepsilon_0$ with $\beta < \varepsilon_0$ is impossible by $\alpha \leq \beta$. Each live case closes by rfl from the MachinePhase join definition.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: epsilon_zero_snap_canonical

$\exists \phi : \text{Ordinal} \rightarrow \text{MachinePhase},$

$(\forall \alpha \beta, \alpha \leq \beta \rightarrow \text{join} (\phi \alpha) (\phi \beta) = \phi \beta) \wedge [\text{hmono}]$

$(\forall n : \mathbb{N}, \phi (\text{fundamentalSeq } n) = c_0) \wedge [\text{h0}]$

$(\forall \alpha, \omega^\alpha = \alpha \rightarrow \phi \alpha = c_1) \wedge [\text{hfp}]$

$\phi \varepsilon_0 = c_1 \wedge [\text{snap}]$

$\forall \alpha, \phi \alpha = c_1 \rightarrow \varepsilon_0 \leq \alpha$ [minimality]

All five conditions verified for the explicit witness $\phi \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$, with no free hypotheses.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: snap_zp2_correspondence

Four independent facts about the same tower sequence:

(i) $\forall n : \mathbb{N}, \text{fundamentalSeq } n < \varepsilon_0$

(ii) $\forall n : \mathbb{N}, \phi (\text{fundamentalSeq } n) = c_0$ where ϕ is the canonical map

(iii) $\text{Filter.Tendsto} (\text{fun } n \mapsto \text{cnfToZp2} (\text{towerNONote } n)) \text{Filter.atTop} (\text{nhds } 0)$

(iv) $\phi \varepsilon_0 = c_1$

The same tower sequence witnesses both the ordinal approach to ε_0 and the 2-adic approach to 0. The full structural identification ($\varepsilon_0 \leftrightarrow \perp$ via a type bridge) remains outside Lean scope — see §V.

Proof: $\langle \text{epsilonZero_tower_lt}, \text{fun } n \mapsto \text{if_pos } \dots, \text{tower_converges_to_zero}, \text{if_neg } (\text{lt_irrefl } \varepsilon_0) \rangle$

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Remaining Formal Gap

The Remaining Gap

The identification of ZPE's MachinePhase element c_1 with the ordinal ϵ_0 via a formal type bridge remains outside Lean scope.

What is proved: a canonical map $\text{Ordinal} \rightarrow \text{MachinePhase}$ assigns c_1 exactly at ϵ_0 and nowhere earlier. What is not proved: a canonical ZPSemilattice morphism $\text{MachinePhase} \rightarrow \mathbb{Z}_2$ that would connect ZPE's $\perp = c_0$ to ZPB's $\perp = 0$ formally.

Such a bridge would require:

(1) $\text{snapEmbed} : \text{MachinePhase} \rightarrow \mathbb{Z}_2$ mapping $c_0 \mapsto 0, c_1 \mapsto 1$

(2) proof that snapEmbed is join-preserving

(3) a bridge theorem deriving hfp from $\text{tower_converges_to_zero}$ via snapEmbed

This would make hfp a theorem rather than a hypothesis in $\text{snap_exactly_at_epsilon_zero}$. The canonical witness ($\text{epsilon_zero_snap_canonical}$) satisfies all five conditions without this bridge; the bridge would close the gap between the two formal instances of $\perp$ across ZPE and ZPB.

Theorem Summary

Theorem	Section	Status
roger_fixed_point_stability	§II	Proved ✓
epsilonZero_fixedPoint	§III	Proved ✓
epsilonZero_eq_nfp	§III	Proved ✓
epsilonZero_eq_iSup	§III	Proved ✓
epsilonZero_tower_lt	§III	Proved ✓
epsilonZero_le_fixedPoint	§III	Proved ✓
fundamentalSeq_strictMono	§III	Proved ✓
tower_stage_zero / _one / _two	§III	Proved ✓
towerNONote_repr	§IV	Proved ✓
cnfToZp2_tower_valuation	§IV	Proved ✓
cnfToZp2_valuation_unbounded	§IV	Proved ✓
fundamentalSeq_zp2_converges	§IV	Proved ✓
tower_converges_to_zero	§IV	Proved ✓
zpe_snap_ordinal_correspondence	§V	Proved ✓

Theorem	Section	Status
epsilonZero_tower_bound	§V	Proved ✓
c1_epsilon_zero_identification	§V	Proved ✓
fundamentalSeq_cofinal	§V	Proved ✓
snap_threshold_is_epsilon_zero	§V	Proved ✓
snap_exactly_at_epsilon_zero	§VI	Proved ✓
kleene_ordinal_snap_bridge	§VI	Proved ✓
snap_map_mono	§VII	Proved ✓
epsilon_zero_snap_canonical	§VII	Proved ✓
snap_zp2_correspondence	§VII	Proved ✓

Axiom Purity

All 23 theorems carry axiom footprint: [propext, Classical.choice, Quot.sound].

These are standard Mathlib infrastructure axioms (ordinal theory, p-adic analysis, computability). They are not ZP-L commitments.

Classical.choice is load-bearing: it is the formal non-constructivity appearing at the diagonal step in each ZP layer (§I). Its presence is expected and documented, not incidental.

Zero sorry in ZPL.lean. Verified: lake build, May 2026.